

Day 5 - ML-Bootcamp

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Probability:-

Terminology:-

① Random Experiment:-

An experiment is called

Random experiment if it satisfied the following two conditions:-

- i) It has more than one possible outcome.
- ii) It is not possible to predict the outcome in advance.

Example:- Toss, rolling a dice.

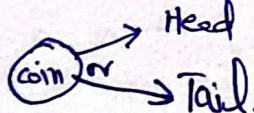
② Trial:-

Trial refers to a single execution of a random experiment. Each trial produces an outcome.

Example:-  { single toss mean single trial. }

③ Outcome:-

Outcome refers to single possible result of a trial.

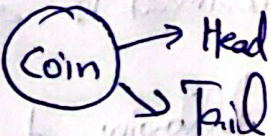
Example:- 

{ Head is a single outcome
Tail is also an outcome }

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④ Sample Space :-

Sample space of a random experiment is the set of all possible outcomes. Generally, one random experiment will have one set of sample space.

Example: 

$$S = \{\text{Head, Tail}\}$$

⑤ Event :-

Event is the specific set of outcomes.

The outcome you are going for is an event.

Example: If you want to predict that, if you roll a die then probability of getting number < 3 .

In this case the event is $\{1, 2\}$

one random experiment can have multiple events.

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Types of Events

① Simple Event:-

It contains exactly the one outcome.

Example:

rolling a die and getting 3 is a simple event.

② Compound Event:-

Event consisting of Compound (two or more) events.

Example:

Getting odd number in rolling a die.

③ Independent Events:

One event can't effect other one.

Example: Rolling a die with a coin. Both are Independent.

④ Dependent Events:-

If others effect the outcome.

Example:- Draw 1 card from 52 cards, then it effect on the next outcome if you don't place it again in the stack.

⑤ Mutually Exclusive Events:-

(or disjoint) if they can't both occur at the same time. Two events are mutually exclusive

Example:- roll a die and expect 2 and 4 both.

⑥ Exhaustive Events:-

A set of events is exhaustive if at least one of the events must occur when the experiment is performed.

Example:- when you roll a die it must be an odd or even number.

⑦ Impossible event:-

A event which can't be occur at any cost.

Example:-

rolling a die and expecting 7.

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What is Probability :-

It is the measure of likelihood that a particular event will occur. It is the fundamental concept in statistics and is used to make predictions and informed decisions.

It is between 0 and 1.

0 — event not happens,

1 — event will certainly happen.

0.5 — 50% — 50%

Empirical Probability vs Theoretical Probability

Empirical Probability (Experimental Probability) :-

It is the Probability measure that is based on observed data, rather than theoretical data (assumptions). It's calculated as the ratio of the number of times a particular event occurs to the total number of trials.

A: Suppose that in our 100 tosses, we get heads 55 times and tails 45 times. What is the empirical probability of getting a head?

Solution:—

$$P(H) = \frac{\text{Totals Heads} \rightarrow 55}{\text{Trails} \rightarrow 100} = \frac{55}{100} \dots$$

B: Let's say you have a bag with 50 marbles. Out of these 50 marbles, 20 are red, 15 are blue, and 15 are green. You start to draw marbles one at a time, replacing the marble back into the bag after each draw. After 200 draws, you find that you've drawn a red marble 80 times, a blue marble 70 times, and a green marble 50 times. What is the empirical probability of getting a red marble.

$$\text{Solution} = \frac{80}{200} \dots$$

Theoretical Probability:-

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* When the trials (number) increases then the empirical probability is close to the theoretical probability.

Theoretical or Classical Probability is used when each outcome in a sample space is equally likely to occur.

Formula:-

Theoretical probability of Event A = $\frac{\text{Number of favorable outcomes}}{\text{Total outcomes in sample space}}$

A:- Consider a tossing of coin 3 times. Find the probability of getting exactly 2 heads.

SS :- {
HHH
HHT
HTH
HTT
THH
THT
TTH
TTT
}

~~P(2H) = 2/8~~

Total outcomes = 2^3

favorable outcome = 3
Total outcomes = 8

* 2 = Head or tail :: 3 = times

$$\frac{3}{8} = 0.375 \text{ or } 37.5\%$$

B:- Consider a rolling 2 dice. What is the probability of getting a sum = 7.

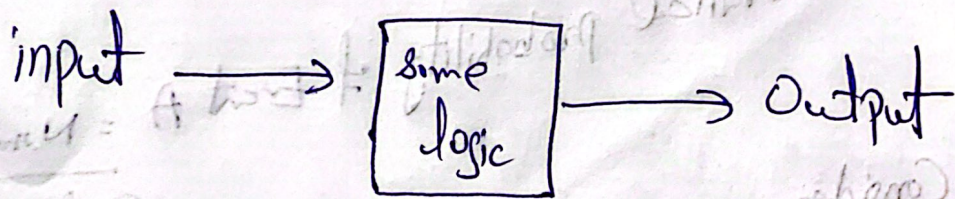
favorable outcome:-

Total outcomes = $6 \times 6 = 36$

$$\frac{6}{36} = \frac{1}{6} \approx 0.1667 \text{ or } 16.67\%$$

Random Variable ^(function) :-

It is a function, that maps the outcomes of a random process (known as the sample space) to a set of real numbers.



Random variable

Discrete

Continuous

Probability Distribution ^(function) :-

rolling die: $SS = \{1, 2, 3, 4, 5, 6\}$

X	1	2	3	4	5	6
P(X)	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$